

PROMPT FOR DETERMINING THE HADWIGER–NELSON NUMBER

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ABSTRACT. This document contains a full frontier-model research prompt for determining the chromatic number of the Euclidean plane when points at unit distance must receive different colors. The exact value is currently one of 5, 6, or 7. Status snapshot: 20 July 2026.

1. Prompt

Current task statement

Let $G_{\mathbb{R}^2}$ be the graph whose vertex set is the Euclidean plane and whose edges join pairs of points at distance exactly 1. The Hadwiger–Nelson problem asks for its chromatic number

$$\chi(\mathbb{R}^2) = \chi(G_{\mathbb{R}^2}).$$

Resolve the problem completely. The current rigorous range is

$$5 \leq \chi(\mathbb{R}^2) \leq 7,$$

so the target value is 5, 6, or 7. Colorings are unrestricted unless explicitly proved otherwise; do not replace the ordinary chromatic number by only the measurable, Borel, periodic, or fractional variant.

Complete-resolution standard

A complete proof of $\chi(\mathbb{R}^2) = k$ must establish both:

- a lower bound $\chi(\mathbb{R}^2) \geq k$, preferably through a finite unit-distance graph with chromatic number at least k and exact algebraic coordinates;
- an upper bound $\chi(\mathbb{R}^2) \leq k$, through an explicit coloring rule assigning every point of the entire plane one of k colors and a rigorous proof that no unit-distance pair is monochromatic;
- exact treatment of all geometric boundaries, exceptional points, tile interfaces, and algebraic distance claims;
- for computational subproofs, independently checkable graph-coloring and geometric certificates.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- a finite graph requiring k colors without an exact proof that all designated edges have Euclidean length 1;
- a numerical coordinate list whose unit distances hold only within tolerance;
- a k -coloring of a lattice, bounded region, dense subset, or almost-everywhere subset rather than all of $\mathbb{R}^2$;
- a measurable or Borel result presented as the unrestricted chromatic number without a proof of equivalence;
- an upper coloring with ambiguous boundary conventions or points assigned no color or more than one color;
- an improvement that leaves two possible integer values.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- generate finite unit-distance graphs from triangular, hexagonal, cyclotomic, and algebraic point sets, with exact coordinates and SAT-certified chromatic number;
- compose and glue known high-chromatic unit-distance gadgets while controlling accidental coincidences and preserving exact embeddability;
- use symmetry, group orbits, algebraic-number fields, graph products, Minkowski sums, and constrained geometric optimization to discover compact witnesses;
- construct periodic or quasiperiodic plane colorings from lattices, tilings, Voronoi cells, strips, and multi-scale hierarchical rules;
- convert upper-coloring verification into exact distance-avoidance inequalities between closed, open, or half-open color regions;
- apply SAT, SMT over algebraic reals, interval arithmetic, cylindrical algebraic decomposition, and proof assistants to certify both graph and geometry;
- investigate topological, harmonic-analytic, density, and additive-combinatorial obstructions to 5- or 6-colorings;
- search for a finite obstruction to $(k-1)$ -colorability and, independently, a global constructive coloring with k colors for each remaining k .

Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- for a lower-bound graph, verify every claimed edge has squared distance exactly 1 in an explicitly specified number field;
- validate non- $(k-1)$ -colorability with a checkable SAT proof and a second independent graph-coloring

implementation;

- ensure the lower-bound graph is a simple graph and that coincident coordinate expressions do not collapse distinct vertices;
- for an upper coloring, prove every point receives exactly one color, including all tile edges, vertices, limit points, and exceptional sets;
- compute the full set of displacement vectors between same-colored regions and prove none has norm 1;
- distinguish exactly-unit constraints from less-than-or-equal-to-unit packing constraints;
- audit any use of compactness or the de Bruijn-Erdos theorem for its set-theoretic and finite-subgraph hypotheses.

Required final artifact

The final response must package both sides of the exact equality:

- an exact lower-bound witness, including coordinate definitions, edge list, and a chromatic-number certificate;
- a complete upper-coloring specification in mathematical and machine-readable form;
- a verifier for all finite graph and algebraic-distance claims;
- a formal or rigorously checkable boundary and distance analysis for the global coloring;
- a manifest and hashes for all data, source code, proof certificates, and rendered diagrams.

Public search may be used for standard unit-distance graph constructions, known 5-chromatic witnesses, and the classical 7-coloring. Do not search for a purported new final value or answer by restating the current interval.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.